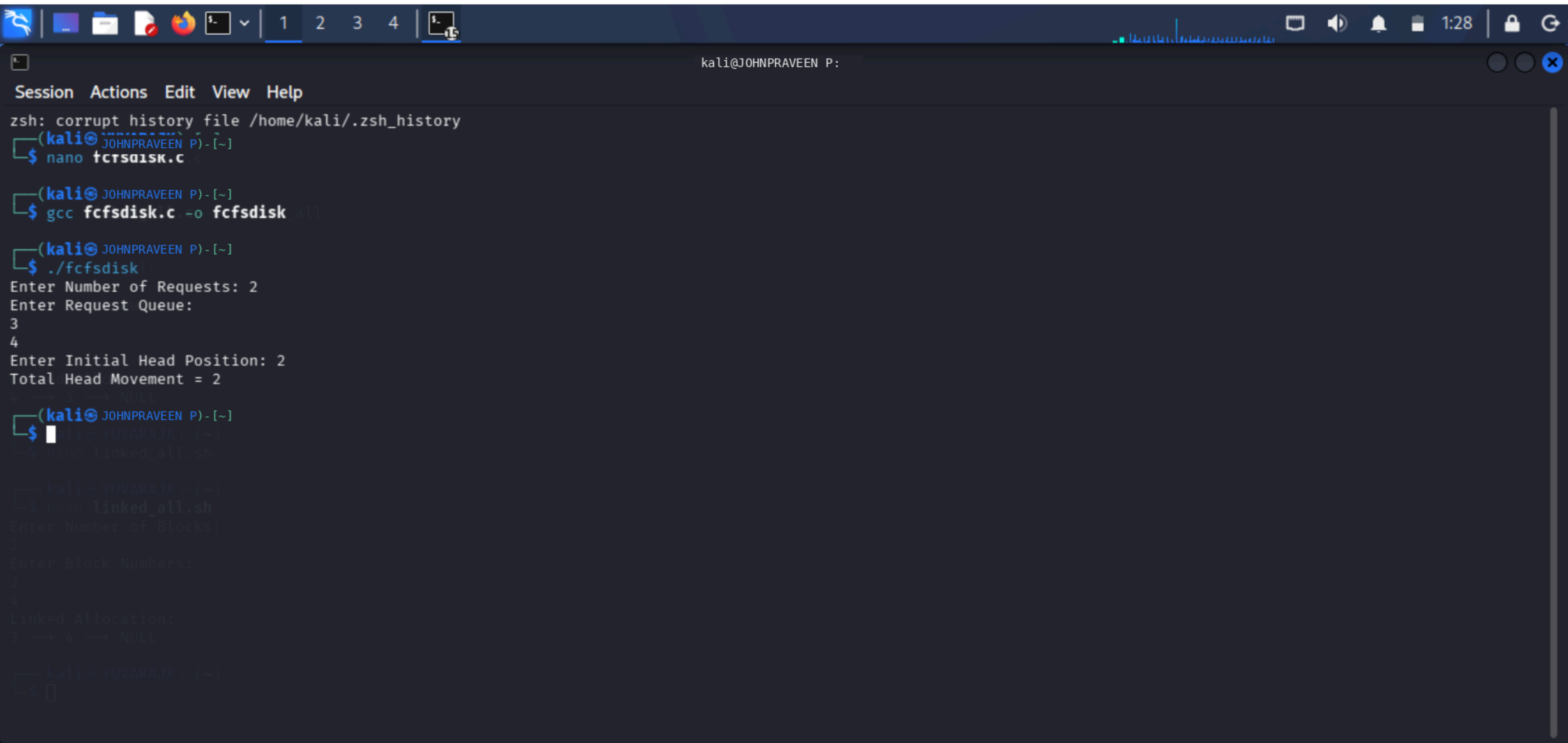


C PROGRAMMING :FCFS DISK



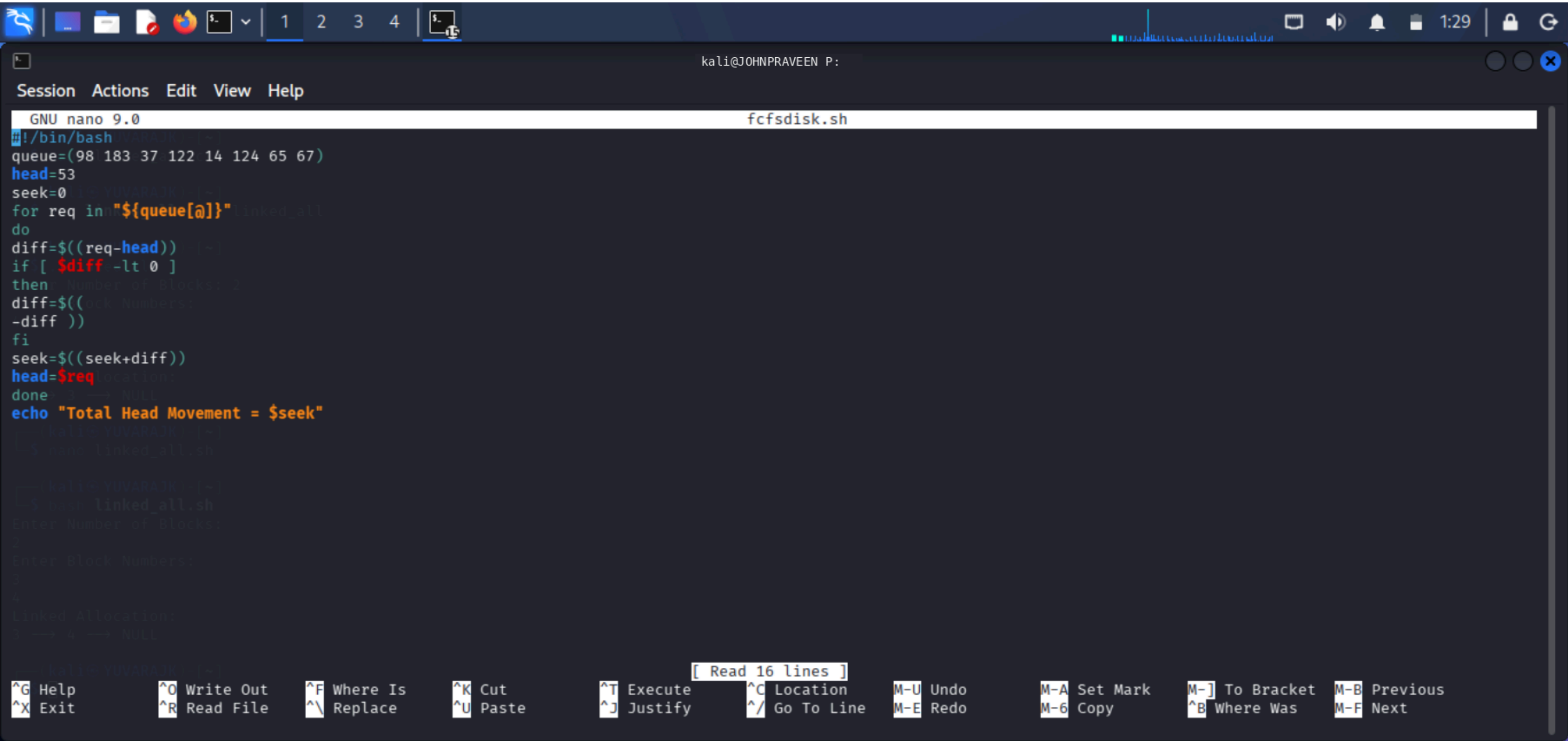
```
GNU nano 9.0 fcfsdisk.c
#include<stdio.h>
#include<stdlib.h>
int main()
{
    int req[20], n, head, i;
    int seek = 0;
    printf("Enter Number of Requests: ");
    scanf("%d",&n);
    printf("Enter Request Queue:\n");
    for(i=0;i<n;i++)
    scanf("%d",&req[i]);
    printf("Enter Initial Head Position: ");
    scanf("%d",&head);
    for(i=0;i<n;i++)
    {
        seek += abs(req[i] - head);
        head = req[i];
    }
    printf("Total Head Movement = %d\n",seek);
    return 0;
}
```

OUTPUT :



```
zsh: corrupt history file /home/kali/.zsh_history
(kali@JOHNRAVEEN P)~]
$ nano fcfsdisk.c
(kali@JOHNRAVEEN P)~]
$ gcc fcfsdisk.c -o fcfsdisk
(kali@JOHNRAVEEN P)~]
$ ./fcfsdisk
Enter Number of Requests: 2
Enter Request Queue:
3
4
Enter Initial Head Position: 2
Total Head Movement = 2
(kali@JOHNRAVEEN P)~]
$
```

SHELL PROGRAMMING : FCFS DISK

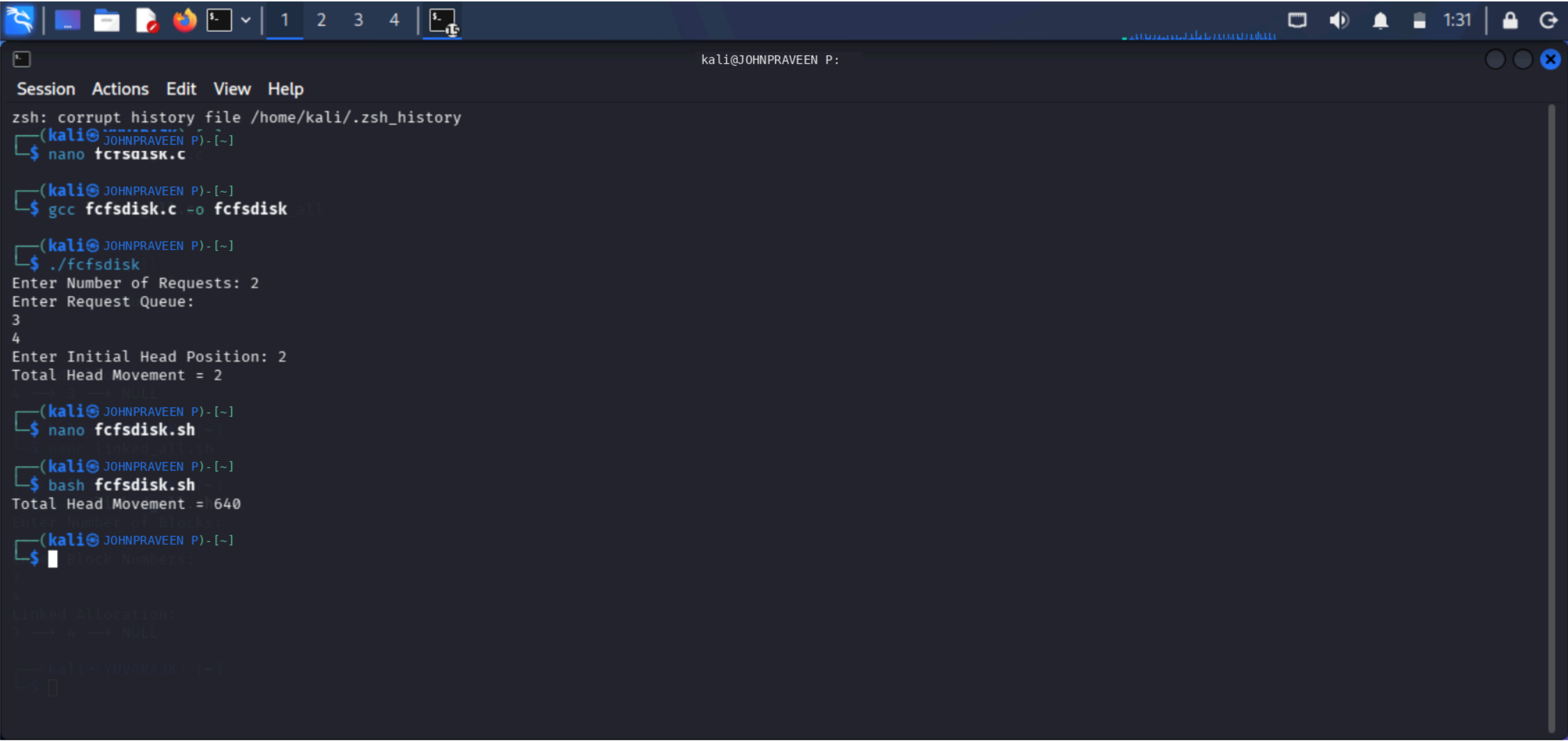


The screenshot shows a terminal window with the nano text editor open, editing a file named `fcfsdisk.sh`. The script is a shell program that implements the First-Come, First-Served (FCFS) disk scheduling algorithm. It takes a queue of requests and an initial head position as input and calculates the total head movement. The script is as follows:

```
#!/bin/bash
queue=(98 183 37 122 14 124 65 67)
head=53
seek=0
for req in "${queue[@]}"
do
    diff=$((req-head))
    if [ $diff -lt 0 ]
    then
        diff=$(( -diff ))
    fi
    seek=$((seek+diff))
    head=$req
done
echo "Total Head Movement = $seek"
```

The nano editor interface includes a menu bar with options like Session, Actions, Edit, View, and Help. The status bar at the bottom shows various keyboard shortcuts for editing and navigation.

OUTPUT :



The screenshot shows a terminal window with the following commands and output:

```
zsh: corrupt history file /home/kali/.zsh_history
(kali@JOHNRAVEEN P)~$ nano fcfsdisk.c
(kali@JOHNRAVEEN P)~$ gcc fcfsdisk.c -o fcfsdisk
(kali@JOHNRAVEEN P)~$ ./fcfsdisk
Enter Number of Requests: 2
Enter Request Queue:
3
4
Enter Initial Head Position: 2
Total Head Movement = 2
(kali@JOHNRAVEEN P)~$ nano fcfsdisk.sh
(kali@JOHNRAVEEN P)~$ bash fcfsdisk.sh
Total Head Movement = 640
(kali@JOHNRAVEEN P)~$
```

The output demonstrates the execution of the `fcfsdisk` program and the `fcfsdisk.sh` script. The first run shows a total head movement of 2 for a queue of requests [3, 4] and an initial head position of 2. The second run shows a total head movement of 640 for the same queue and head position.

C PROGRAMMING : SSTF DISK

```
GNU nano 9.0 sstfdisk.c *
#include<stdio.h>
#include<stdlib.h>
int main() {
int req[20], visited[20]={0};
int n, head, i, count=0;
int seek=0, index, min, distance;
printf("Enter Number of Requests: ");
scanf("%d",&n);
printf("Enter Request Queue:\n");
for(i=0;i<n;i++)
scanf("%d",&req[i]);
printf("Enter Initial Head Position: ");
scanf("%d",&head);
while(count<n){
min=9999;
for(i=0;i<n;i++){
if(!visited[i]){
distance=abs(req[i]-head);
if(distance<min){
min=distance;
index=i;
}}
seek+=min;
head=req[index];
visited[index]=1;
count++;
}
printf("Total Head Movement = %d\n",seek);
return 0;
}
```

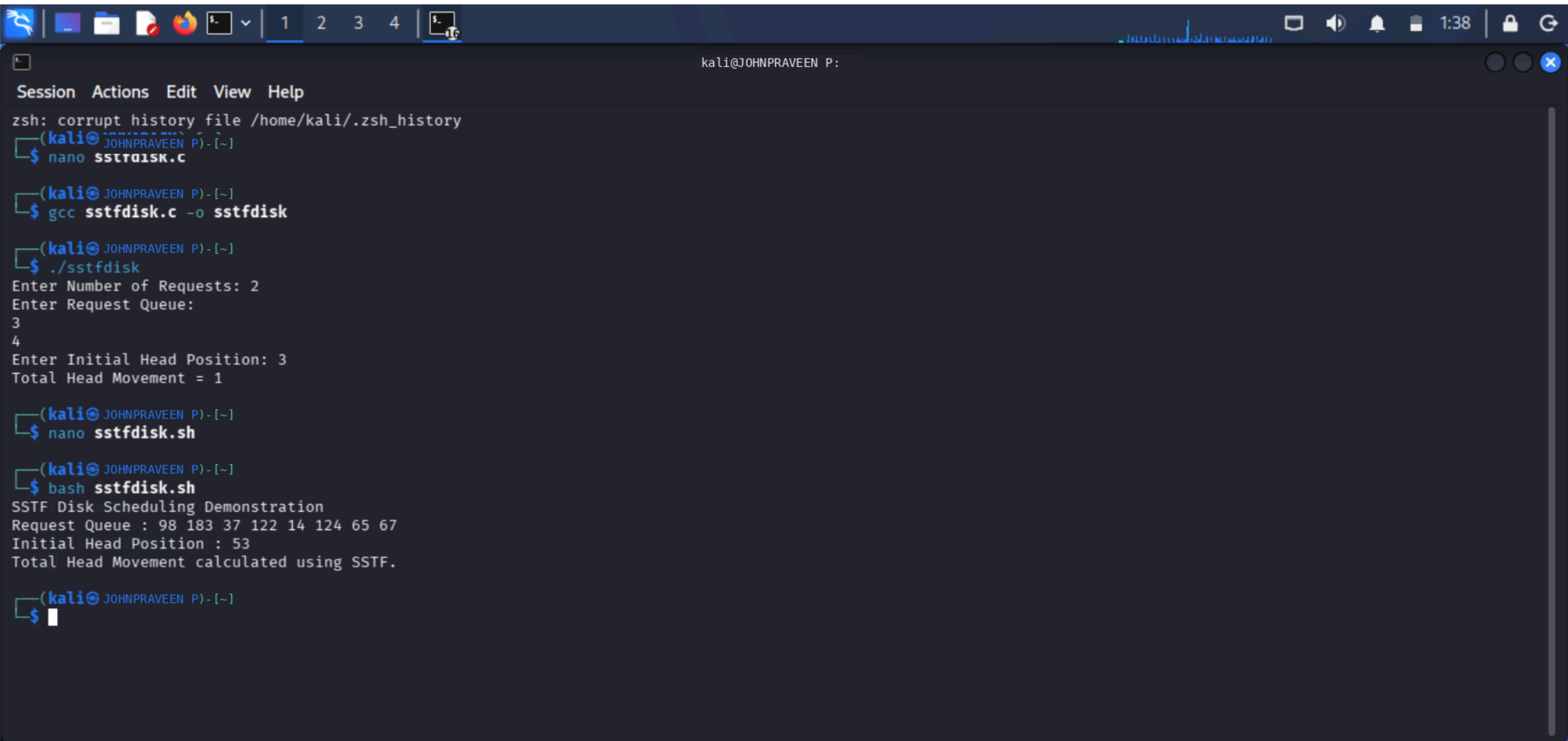
OUTPUT :

```
zsh: corrupt history file /home/kali/.zsh_history
(kali@JOHNPRAVEEN P) ~
$ nano sstfdisk.c
(kali@JOHNPRAVEEN P) ~
$ gcc sstfdisk.c -o sstfdisk
(kali@JOHNPRAVEEN P) ~
$ ./sstfdisk
Enter Number of Requests: 2
Enter Request Queue:
3
4
Enter Initial Head Position: 3
Total Head Movement = 1
(kali@JOHNPRAVEEN P) ~
$ ./sstfdisk.sh
Enter Number of Requests: 2
Enter Request Queue:
3
4
Enter Initial Head Position: 3
Total Head Movement = 1
(kali@JOHNPRAVEEN P) ~
$
```

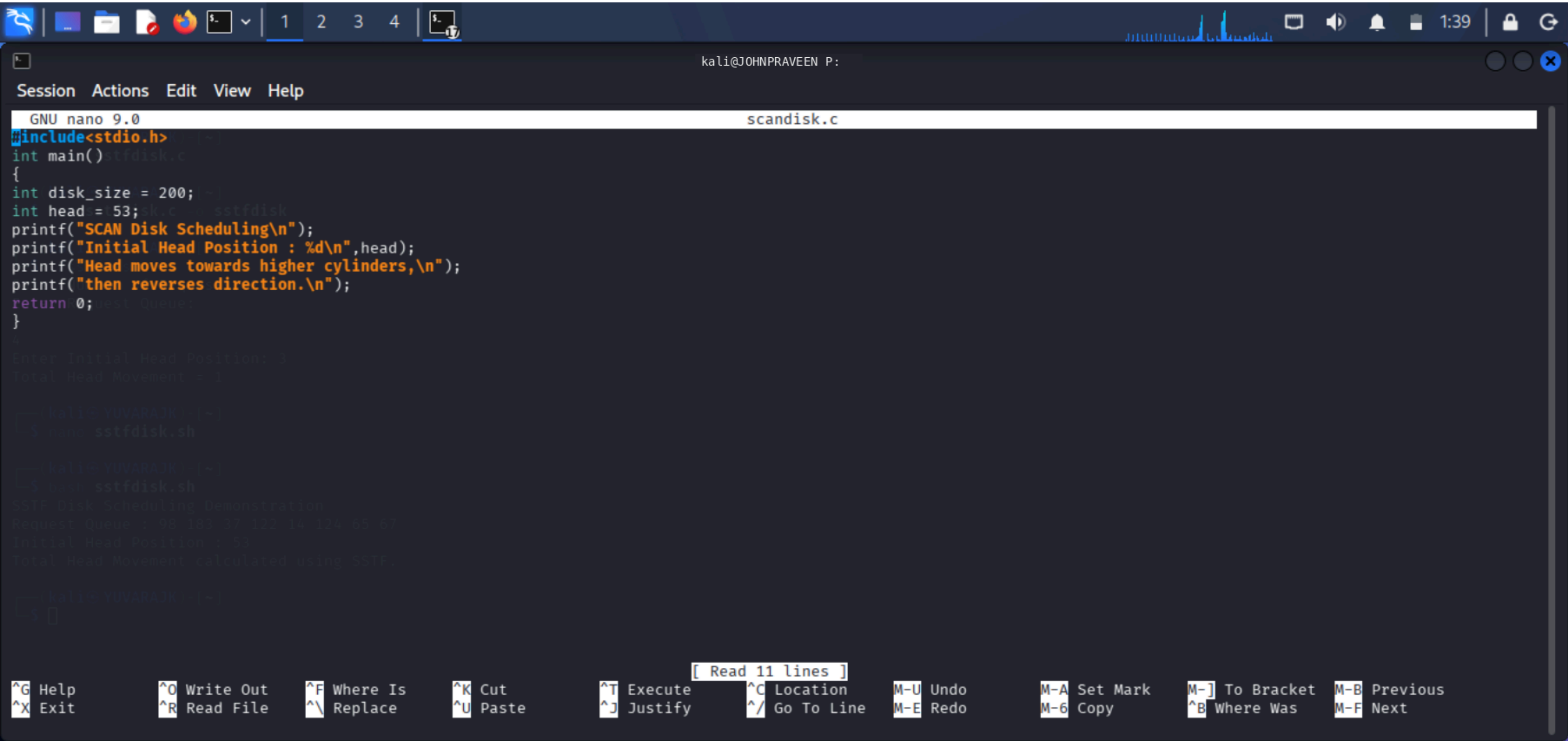
SHELL PROGRAMMING : SSTF DISK



OUTPUT :



C PROGRAMMING : SCAN DISK



```
GNU nano 9.0 scandisk.c
#include<stdio.h>
int main()
{
    int disk_size = 200;
    int head = 53;
    printf("SCAN Disk Scheduling\n");
    printf("Initial Head Position : %d\n",head);
    printf("Head moves towards higher cylinders,\n");
    printf("then reverses direction.\n");
    return 0;
}
```

Enter Initial Head Position : 53
Total Head Movement : 1

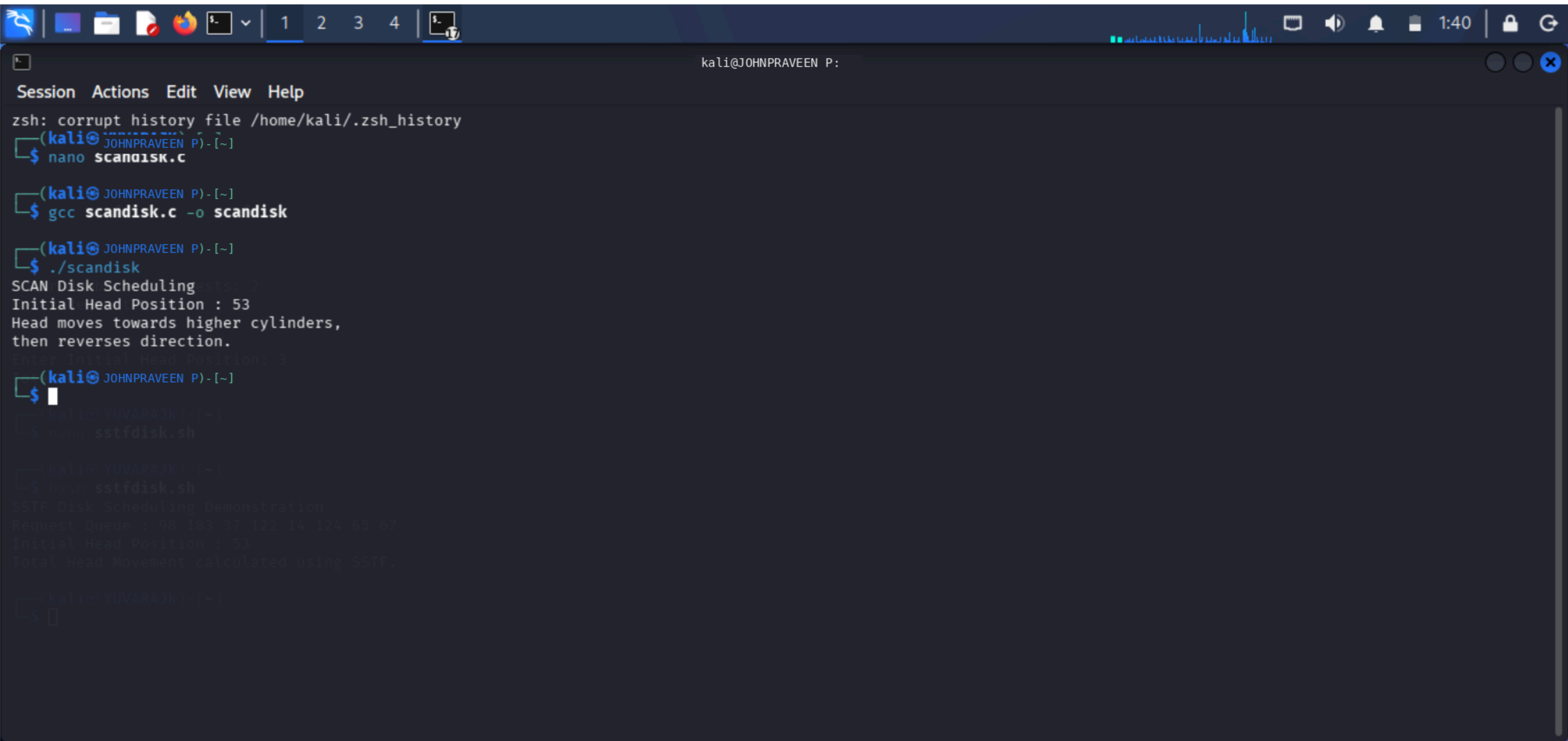
kali@JOHNRAVEEN P:~\$./scandisk.sh
kali@JOHNRAVEEN P:~\$./scandisk.sh
kali@JOHNRAVEEN P:~\$./scandisk.sh
kali@JOHNRAVEEN P:~\$./scandisk.sh
SCAN Disk Scheduling Demonstration
Request queue : 20 10 12 14 124 45 40
Initial Head Position : 53
Total Head Movement calculated using SCAN : 1

kali@JOHNRAVEEN P:~\$
[0]

[Read 11 lines]

^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute ^C Location M-U Undo M-A Set Mark M-I To Bracket M-B Previous
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^_ Go To Line M-E Redo M-G Copy ^B Where Was M-F Next

OUTPUT :



```
zsh: corrupt history file /home/kali/.zsh_history
(kali@JOHNRAVEEN P:~)
$ nano scandisk.c

(kali@JOHNRAVEEN P:~)
$ gcc scandisk.c -o scandisk

(kali@JOHNRAVEEN P:~)
$ ./scandisk
SCAN Disk Scheduling
Initial Head Position : 53
Head moves towards higher cylinders,
then reverses direction.

(kali@JOHNRAVEEN P:~)
$
```

SHELL PROGRAMMING : SCAN DISK

The image shows a terminal window on a Kali Linux system. The window title is 'kali@JOHNPRAVEEN P:'. The terminal is running a script named 'scandisk.sh' using the 'nano' text editor. The script's output is as follows:

```
GNU nano 9.0 scandisk.sh
#!/bin/bash
echo "SCAN Disk Scheduling"
echo "Head moves in one direction and then reverses."

# Disk layout
disk=(10 20 30 40 50 60 70 80 90)

# Request sequence
requests=(10 20 30 40 50 60 70 80 90)

# Initial head position
head=50

# Total head movement
total_movement=0

# SSTF algorithm
while [ ${#requests[@]} -gt 0 ]
do
    # Find the request closest to the current head
    min_diff=999
    for i in $(seq 0 $((${#requests[@]} - 1)))
    do
        diff=$(( ${requests[i]} - $head ))
        diff=$(( ${diff} < 0 ? -${diff} : ${diff} ))
        if [ $diff -lt $min_diff ]
        then
            min_diff=$diff
            closest=$i
        fi
    done

    # Move the head to the closest request
    head=${requests[$closest]}

    # Calculate the head movement for this request
    movement=$(( ${requests[$closest]} - $head ))
    movement=$(( ${movement} < 0 ? -${movement} : ${movement} ))
    total_movement=$(( total_movement + movement ))

    # Remove the request from the sequence
    requests=("${requests[@:closest}" "${requests[@:closest+1]}"*)
done

# Print the total head movement
echo "Total head movement calculated using SSTF: $total_movement"

exit 0
```

The terminal window has a status bar at the bottom with various keyboard shortcuts for nano, such as ^G Help, ^O Write Out, ^F Where Is, ^K Cut, ^T Execute, ^C Location, M-U Undo, M-A Set Mark, M-] To Bracket, M-B Previous, ^X Exit, ^R Read File, ^_ Replace, ^J Paste, ^/_ Go To Line, M-E Redo, M-G Copy, ^B Where Was, and M-F Next. There is also a small notification that says '[Read 3 lines]'.

OUTPUT :

The screenshot shows a Kali Linux terminal window with the following content:

```

kali@JOHNPRAVEEN P:
Session Actions Edit View Help

zsh: corrupt history file /home/kali/.zsh_history
(kali@JOHNPRAVEEN P)-[~]
$ nano scandisk.c

(kali@JOHNPRAVEEN P)-[~]
$ gcc scandisk.c -o scandisk

(kali@JOHNPRAVEEN P)-[~]
$ ./scandisk
SCAN Disk Scheduling
Initial Head Position : 53
Head moves towards higher cylinders,
then reverses direction.

(kali@JOHNPRAVEEN P)-[~]
$ nano scandisk.sh

(kali@JOHNPRAVEEN P)-[~]
$ bash scandisk.sh
SCAN Disk Scheduling
Head moves in one direction and then reverses.

(kali@JOHNPRAVEEN P)-[~]
$

```

C PROGRAMMING : CIRCULAR SCAN

[illegible]

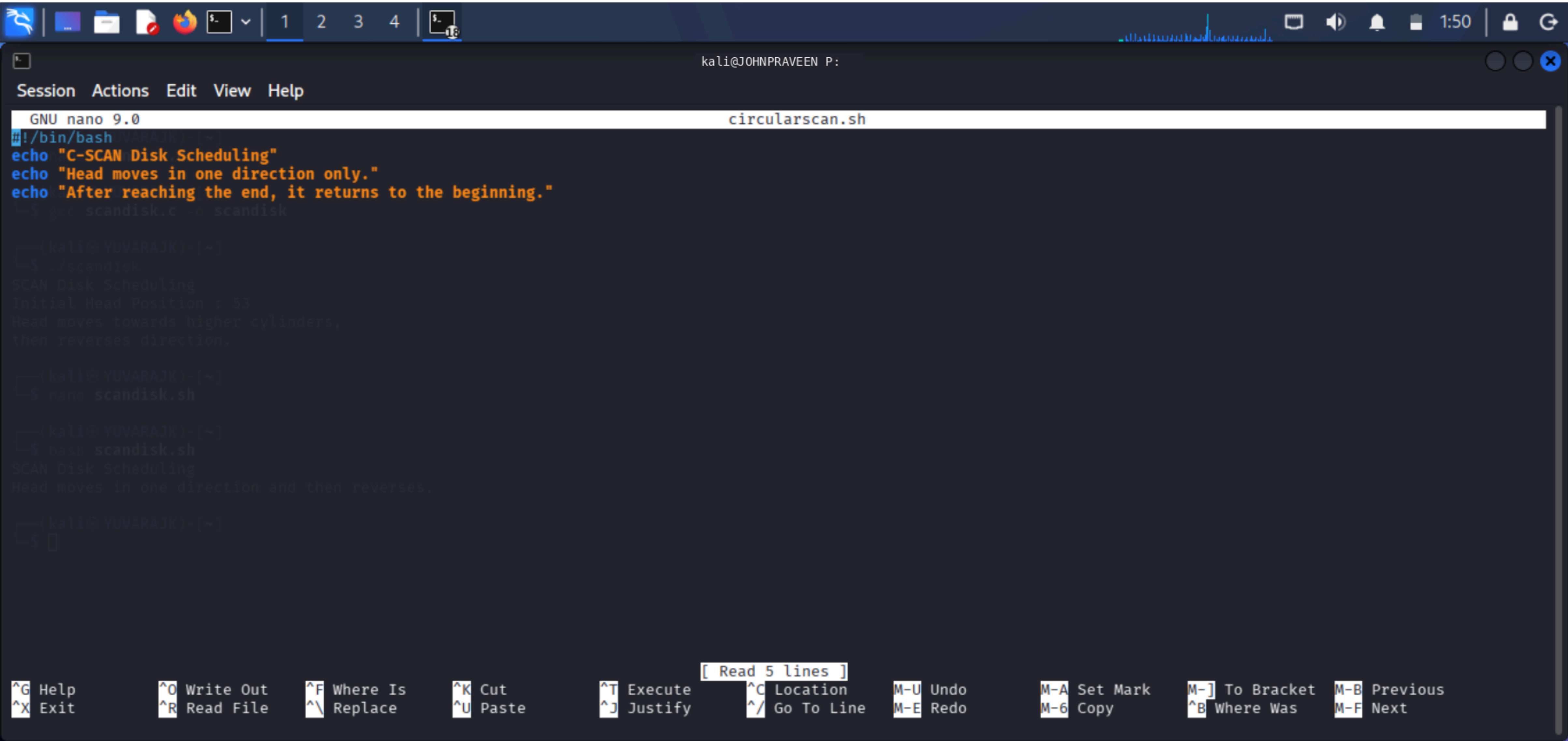
OUTPUT :

```

kali@JOHNPRAVEEN P:
Session Actions Edit View Help
zsh: corrupt history file /home/kali/.zsh_history
(kali@JOHNPRAVEEN P)-[~]
$ nano oscan01sk.c
(kali@JOHNPRAVEEN P)-[~]
$ gcc circularscan.c -o circularscan
(kali@JOHNPRAVEEN P)-[~]
$ ./circularscan
C-SCAN Disk Scheduling
Initial Head Position : 53
Head moves in one direction.
After reaching the end, it returns to the beginning.
(kali@JOHNPRAVEEN P)-[~]
$ ./scandisk.sh
kali@JOHNPRAVEEN P:~$
./scandisk.sh
Disk Disk Scheduling
Head moves in one direction and then reverse.
kali@JOHNPRAVEEN P:~$
$

```

SHELL PROGRAMMING : CIRCULAR SCAN



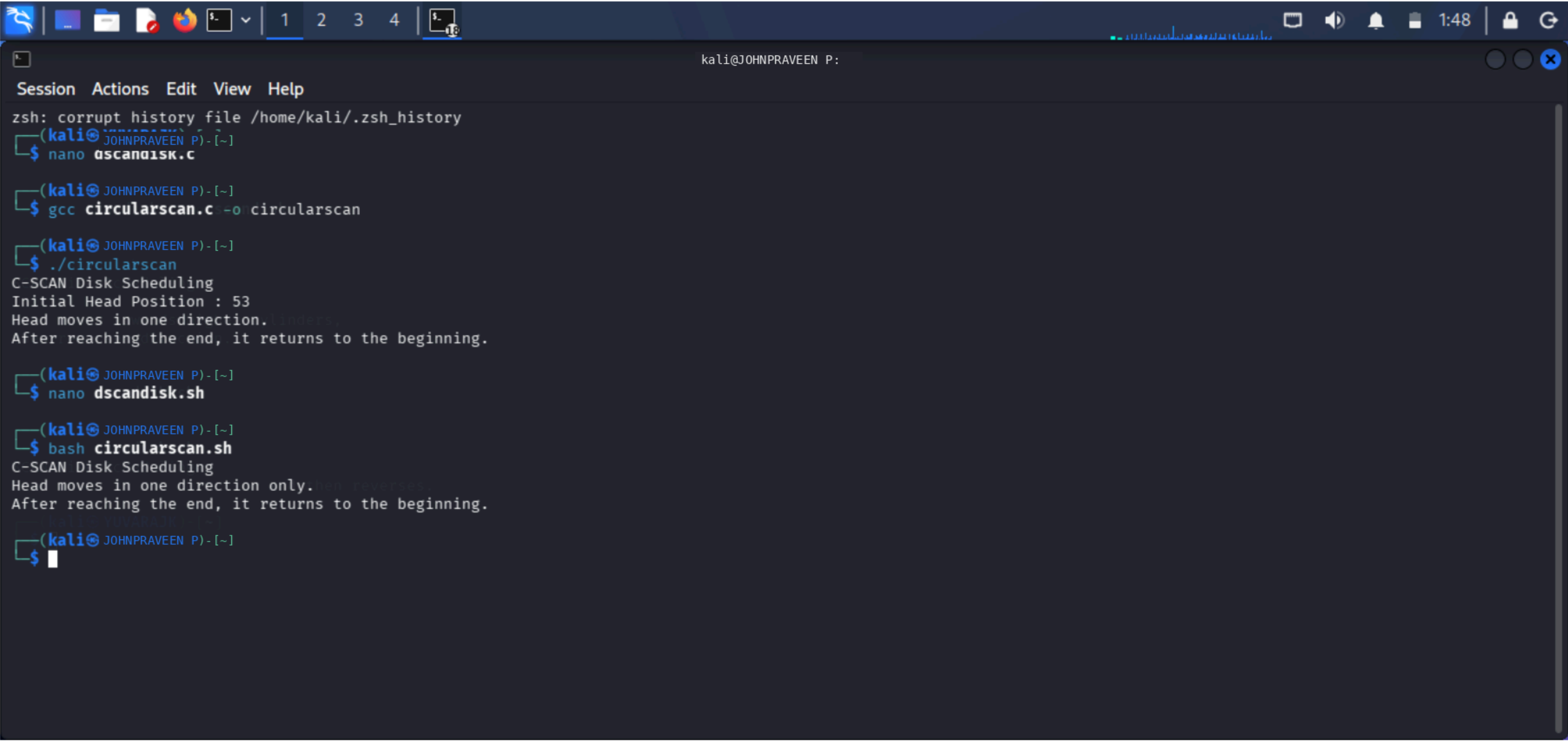
```
GNU nano 9.0 circularscan.sh
#!/bin/bash
echo "C-SCAN Disk Scheduling"
echo "Head moves in one direction only."
echo "After reaching the end, it returns to the beginning."
scandisk.c scandisk

kali@JOHNPRAVEEN P:~$
kali@JOHNPRAVEEN P:~$ cat circularscan.sh
#!/bin/bash
echo "C-SCAN Disk Scheduling"
echo "Head moves in one direction only."
echo "After reaching the end, it returns to the beginning."
scandisk.c scandisk

kali@JOHNPRAVEEN P:~$ cat circularscan.sh
#!/bin/bash
echo "C-SCAN Disk Scheduling"
echo "Head moves in one direction only."
echo "After reaching the end, it returns to the beginning."
scandisk.c scandisk

kali@JOHNPRAVEEN P:~$
```

OUTPUT :



```
zsh: corrupt history file /home/kali/.zsh_history
kali@JOHNPRAVEEN P:~$ nano ascanaisk.c
kali@JOHNPRAVEEN P:~$ gcc circularscan.c -o circularscan
kali@JOHNPRAVEEN P:~$ ./circularscan
C-SCAN Disk Scheduling
Initial Head Position : 53
Head moves in one direction.
After reaching the end, it returns to the beginning.
kali@JOHNPRAVEEN P:~$ nano dscandisk.sh
kali@JOHNPRAVEEN P:~$ bash circularscan.sh
C-SCAN Disk Scheduling
Head moves in one direction only.
After reaching the end, it returns to the beginning.
kali@JOHNPRAVEEN P:~$
```